Image Shrinking

The process opposite to enlarge an image is shrinking it. This is not typically done to examine a ROI more closely but to reduce the amount of data that needs to be processed.

Image shrinking is accomplished by taking groups of pixels that are spatially adjacent and mapping them to one pixel. This can be done in one of ***three ways:***

- **Averaging**
- **Median**
- **Decimation.**

For averaging method, we take all the pixels in each group (for example 2×2 block of pixels) and find the average gray level by summing the values select and divided by 2.

For the second method, median, we sort all the pixels values from lowest to highest and then select the middle value.

The third method decimation, also known as sub sampling, entails simply eliminating some of the data. For example to reduce the image by a factor of two, we simply take every other row and column and delete them.

EXAMPLE

Shrink the following sub image using the three shrinking methods (Averaging, Median, Decimation).

$$\begin{bmatrix} 6 & 8 \\ 1 & 9 \end{bmatrix}$$

I

Averaging	**median**	**decimation**
6+8+1+9=24/4=6	1,6,8,9 = (6+8)/2=7	eliminate the first row and the first column and the result is 9

Chapter Two
Image Analysis

2.1 Image Analysis

Image analysis involves manipulating the image data to determine exactly the information necessary to help solve a computer imaging problem. This analysis is typically part of a larger process, is iterative in nature and allows us to answer application specific equations: Do we need color information? Do we need to transform the image data into the frequency domain? Do we need to segment the image to find object information? What are the important features of the image?

Image analysis is primarily **data reduction** process. As we have seen, images contain enormous amount of data, typically on the order hundreds of kilobytes or even megabytes. Often much of this information is not necessary to solve a specific computer imaging problem, so primary part of the image analysis task is to determine exactly what information is necessary. Image analysis is used both computer vision and image processing.

For computer vision, the end product is typically the extraction of high-level information for computer analysis or manipulation. This high level information may include shape parameter to control a robotics manipulator or color and texture features to help in diagnosis of a skin tumor.

In image processing application, image analysis methods may be used to help determine the type of processing required and the specific parameters needed for that processing. For example, determine the degradation function for an image restoration procedure, developing an enhancement algorithm and determining exactly what information is visually important for image compression methods.

2.2 System Model

The image analysis process can be broken down into three primary stages:

1. Preprocessing.
2. Data Reduction.
3. Features Analysis.

1. Preprocessing:

The preprocessing algorithm, techniques and operators are used to perform initial processing that makes the primary data reduction and analysis task easier. They include operations related to:

- Gray –level or spatial quantization (reducing the number of bits per pixel or the image size).
- Is used to remove noise and eliminate irrelevant, visually unnecessary information, (Noise is unwanted information that can result from the image acquisition process) Finding regions of interest for further processing.
- Performing basic algebraic operation on image.
- Enhancing specific image features.
- Reducing data in resolution and brightness.

Preprocessing is a stage where the requirements are typically obvious and simple, such as removal of artifacts from images or eliminating of image information that is not required for the application. For example, in one application we needed to eliminate borders from the images that have been digitized from film. Another example of preprocessing step involves a robotics gripper that needs to pick and place an object ; for this we reduce a gray-level image to binary (two-valued) image that contains all the information necessary to discern the object ‘s outlines.

2. Data Reduction:

Involves either reducing the data in the spatial domain or transforming it into another domain called the frequency domain, and then extraction features for the analysis process.

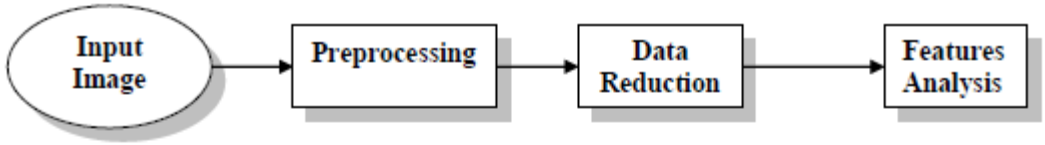
3. Features Analysis:

The features extracted by the data reduction process are examine and evaluated for their use in the application.

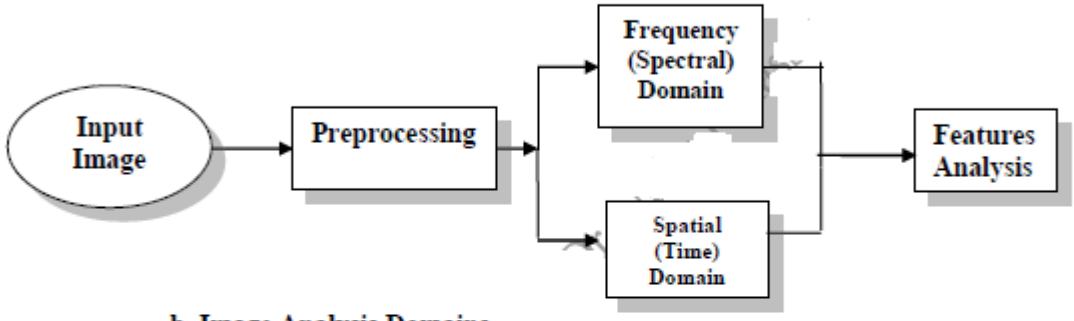
After preprocessing we can perform segmentation on the image in the spatial domain or convert it into the frequency domain via a mathematical transform. After these processes we may choose to filter the image. This filtering process further reduces the data and allows us to extract the feature that we may require for analysis.

Figure (2.1) illustrates the system model of image analysis

Image Analysis System Model



a. Image Analysis



b. Image Analysis Domains

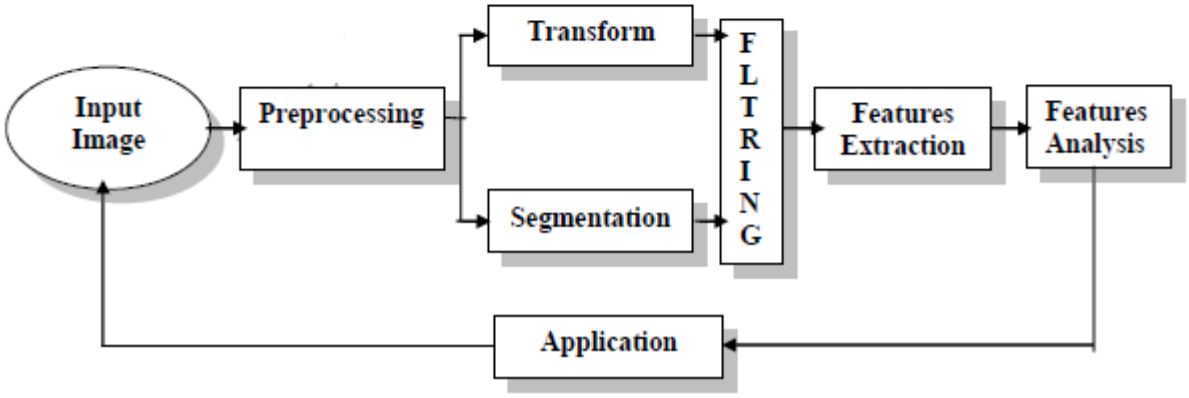


Figure (2.1): System Model of Image Analysis

2.3 Region –of-Interest (ROI)

For image analysis we want to investigate more interested area within the image, called region of interest (ROI). To do this we need operation that modifies the spatial coordinates of the image, and these are categorized as image geometry operations. The image geometry operations discussed here include:

Crop , Zoom, enlarge ,shrink, translate and rotate.

The image crop process is the process of selecting a small portion of the image, a sub image and cutting it away from the rest of the image. After

we have cropped a sub image from the original image we can zoom in on it by enlarge it. The zoom process can be implemented by the following techniques:

1. **Zero-Order Hold.**
2. **First _Order Hold.**
3. **Convolution.**

1. **Zero-Order hold:** is achieved by repeating previous pixel values, thus creating a blocky effect as in the following figure:

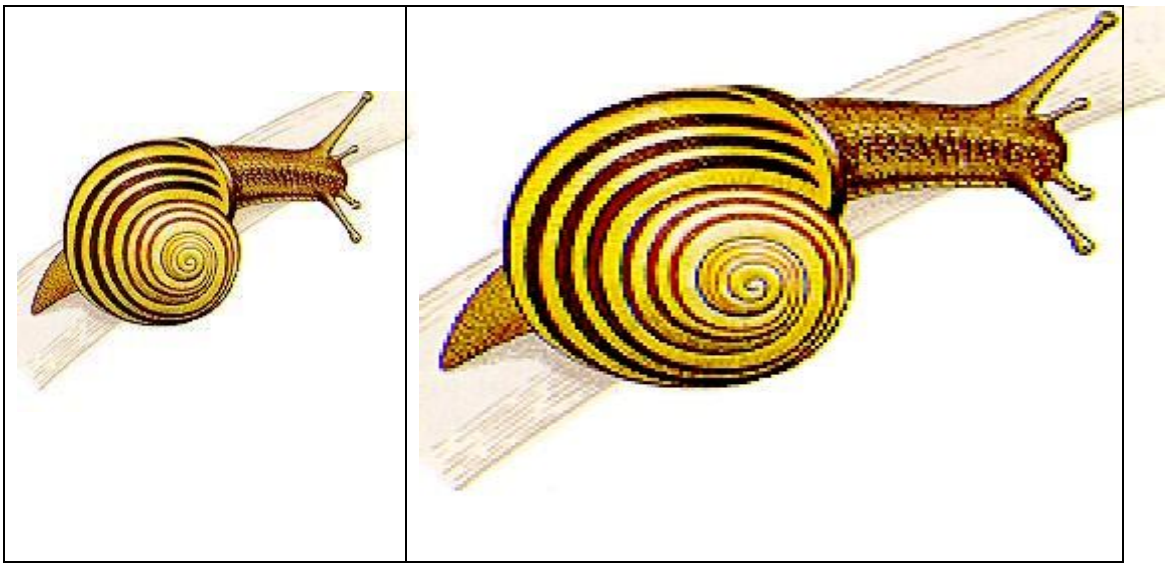


Figure (2.2): Zero Order Hold Method

2.**First _Order Hold:** is performed by finding linear interpolation between a adjacent pixels, i.e., finding the average value between two pixels and use that as the pixel value between those two, we can do this for the rows first as follows:

Original Image Array

$$\begin{bmatrix} 8 & 4 & 8 \\ 4 & 8 & 4 \\ 8 & 2 & 8 \end{bmatrix}$$

Image with Rows Expanded

$$\begin{bmatrix} 8 & 6 & 4 & 6 & 8 \\ 4 & 6 & 8 & 6 & 4 \\ 8 & 5 & 2 & 5 & 8 \end{bmatrix}$$

The first two pixels in the first row are averaged (8+4)/2=6, and this number is inserted between those two pixels. This is done for every pixel pair in each row.

Image with rows and columns expanded

$$\begin{bmatrix} 8 & 6 & 4 & 6 & 8 \\ 6 & 6 & 6 & 6 & 6 \\ 4 & 6 & 8 & 6 & 4 \\ 6 & 5.5 = 6 & 5 & 5.5 = 6 & 6 \\ 8 & 5 & 2 & 5 & 8 \end{bmatrix}$$

This method allows us to enlarge an N×N sized image to a size of (2N-1) × (2N-1) and be repeated as desired.

3- Convolution: this process requires a mathematical process to enlarge an image. This method required two steps:

1. Extend the image by adding rows and columns of zeros between the existing rows and columns.
2. Perform the convolution.

The image is extended as follows**:**

Original Image Array

$$\begin{pmatrix} 3 & 5 & 7 \\ 2 & 7 & 6 \\ 3 & 4 & 9 \end{pmatrix}$$

Image extended with zeros

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 5 & 0 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 7 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 4 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

An image of the dimensions of 2 rows and 2 columns and zoom it twice using First _Order Hold

1	2
3	4

zoom it row wise and then column wise.

1	1.5	2
3	3.5	4

1	1.5	2
2	2.5	3
3	3.5	4

K-Times zooming

Introduction:

K times is the third zooming method we are going to discuss. It is one of the most perfect zooming algorithm discussed so far. It caters the challenges of both twice zooming and pixel replication. K in this zooming algorithm stands for zooming factor.

Working:

It works like this way.

First of all, you have to take two adjacent pixels as you did in the zooming twice. Then you have to subtract the smaller from the greater one. We call this output (OP).

Divide the output(OP) with the zooming factor(K). Now you have to add the result to the smaller value and put the result in between those two values.

Add the value OP again to the value you just put and place it again next to the previous putted value. You have to do it till you place k-1 values in it.

Repeat the same step for all the rows and the columns , and you get a zoomed images.

Example:

Suppose you have an image of 2 rows and 3 columns, which is given below. And you have to zoom it thrice or three times.

15	30	15
30	15	30

K=3

The number of inserted values=K-1=3-1=2

15/3=5

Value1=15+5=20

Value2=Value1+5=20+5=25

15	20	25	30	25	20	15
20	21.6	23.2	25	23.2	21.6	20
25	23.2	21.6	20	21.6	23.2	25
30	25	20	15	20	25	30

The new Image size= 4x7

Image Algebra

There are two primary categories of algebraic operations applied to image:

1. **Arithmetic operations.**
2. **Logic operations.**

Addition, subtraction, division and multiplications comprise the **arithmetic operations**, while AND, OR and NOT makeup the **logic operations**.

These operations which require only one image, and are done on a pixel –by-pixel basis.

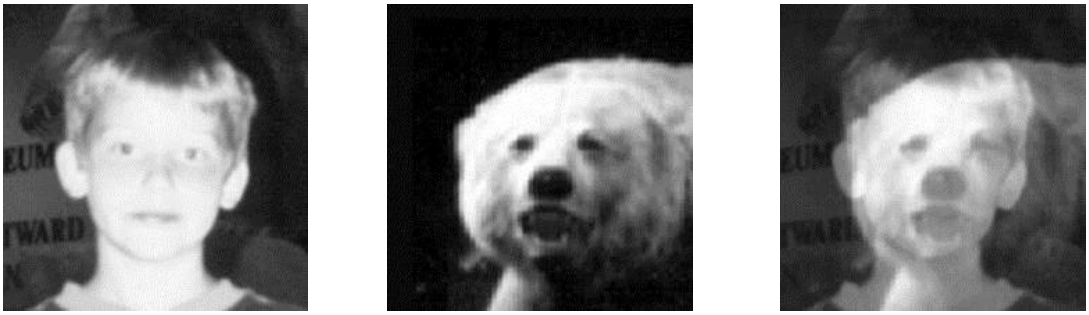
To apply the arithmetic operations on two images, we simply operate on corresponding pixel values, which means that the value of a pixel in the output image depends only on the values of the corresponding pixels in the input images. Hence, the images normally have to be of the same size. For example to add image I_1 and I_2 to create I_3:

$$\underset{I_1}{\begin{pmatrix} 3 & 4 & 7 \\ 3 & 4 & 5 \\ 2 & 4 & 6 \end{pmatrix}} + \underset{I_2}{\begin{pmatrix} 6 & 6 & 6 \\ 4 & 2 & 6 \\ 3 & 5 & 5 \end{pmatrix}} = \underset{I_3}{\begin{pmatrix} 3+6 & 4+6 & 7+6 \\ 3+4 & 4+2 & 5+6 \\ 2+3 & 4+5 & 6+5 \end{pmatrix}} = \begin{pmatrix} 9 & 10 & 13 \\ 7 & 6 & 11 \\ 5 & 9 & 11 \end{pmatrix}$$

- **<u>Addition</u>** is used to combine the information in two images as shown in figure (2.5).Or adding a constant value (scalar) to an image causes an increase (or decrease if the value is less than zero) in its overall brightness. Applications include development of image restoration algorithm for molding additive noise, and special effects, such as image morphing in motion pictures.

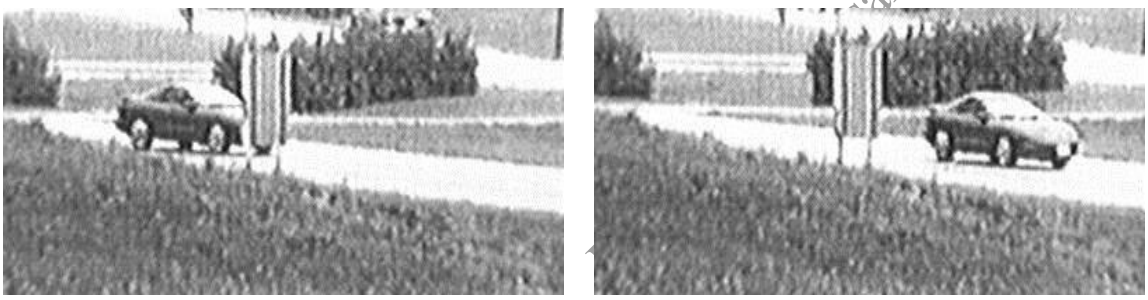
• **Subtraction** of two images is often used to **detect motion**, consider the case where nothing has changed in a sense; the image resulting from subtraction of two sequential image is filled with zero-a black image. If something has moved in the scene, subtraction produces a nonzero result at the location of movement. Subtraction process also used to detect the defects in the images.

Applications include Object tracking, Medical imaging. Subtraction can result in a negative values for certain pixels. When this occurs with unsigned data types, the subtract function truncates the negative value to zero (0), which displays as black

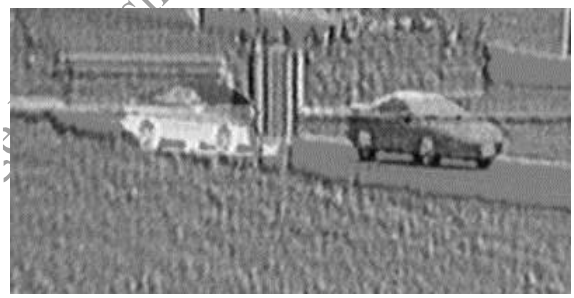


a. First Original image b. Second Original c. Addition of two images

Figure (2.5): Image Addition.



a. Original scene b. Same scene later



c. Subtraction of scene a from scene b

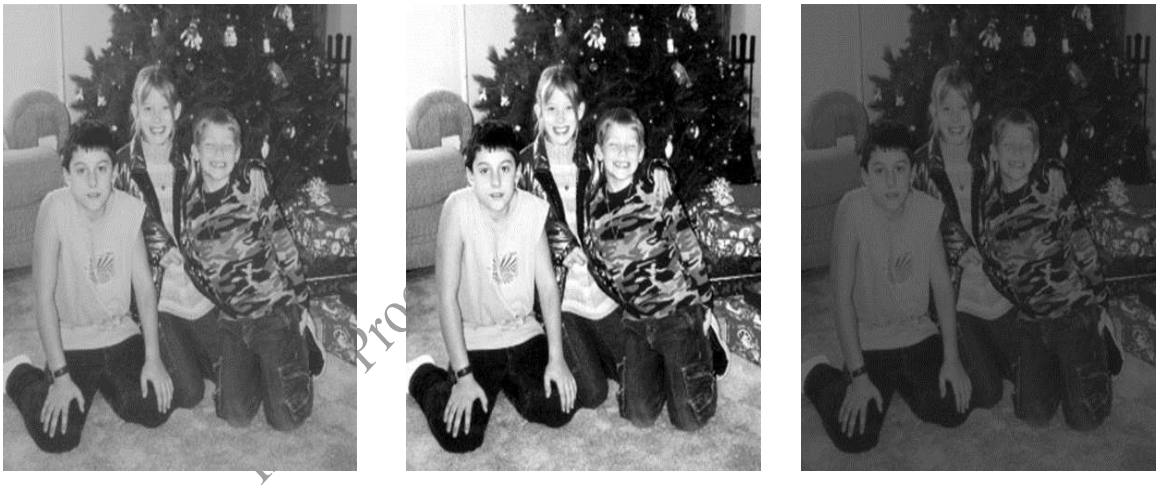
Figure (2.6): Image Subtraction.

- Multiplication and Division are used **to adjust the brightness of an image**. (The Brightness adjustment is often used as a processing step in image enhancement). **Multiplication of the pixel values by a (scalar) smaller than one will darken the image, while multiplication by scalar greater than on will brightness the image**. Multiplication process often used for masking operations, see figure(2.9).



a. Cameraman image **b.**X-ray image of hand c. Multiplication of two images

Figure (2.7): Image Multiplication



a. Original image b. Image divided by value<1 c. Image divided by value >1

Figure (2.8): Image Division



Figure 2.9 : (Image Multiplication by Scalar (smaller than one)

Logical operations apply only ***to binary images***, whereas arithmetic operations apply to multi-valued pixels. Logical operations are basic tools in binary image processing, where they are used for tasks such as ***masking***, ***feature detection***, and ***shape analysis***. Logical operations on entire image are performed pixel–by–pixel. Because the AND operation of two binary variablesis1only when both variables are 1, the result at any location in a resulting AND image is1 only if the corresponding pixels in the two input images are 1. As logical operation involve only one pixel location at a time, they can be done in place, as in the case of arithmetic operations .The XOR (exclusive OR)operationyieldsa1whenoneorotherpixel(but not both)is 1,and it yields a 0 otherwise. The operation is unlike the OR operation, whichis1, when one or the other pixel is1,or both pixels are1.

	AND				OR				XOR			
input1	1	1	0	0	1	1	0	0	1	1	0	0
Input2	1	0	1	0	1	0	1	0	1	0	1	0
output	1	0	0	0	1	1	1	0	0	1	1	0

- Logical AND &OR operations are useful for the ***masking and compositing*** of images For example, if we compute the AND of a binary image with some other image, then pixels for which the corresponding value in the binary image is 1 will be preserved, but pixels for which the corresponding binary value is 0 will be set to 0(erased) Thus the binary image acts as a "***mask***" that ***removes*** information from certain parts of the image.
- On the other hand, if we compute the OR of a binary image with some other image, the pixels for which the corresponding value in the binary image is 0 will be ***preserved***, but pixels for which the corresponding binary value is 1, will be set to 1(cleared).

So, masking is a simple method to extract a region of interest (ROI) from an image

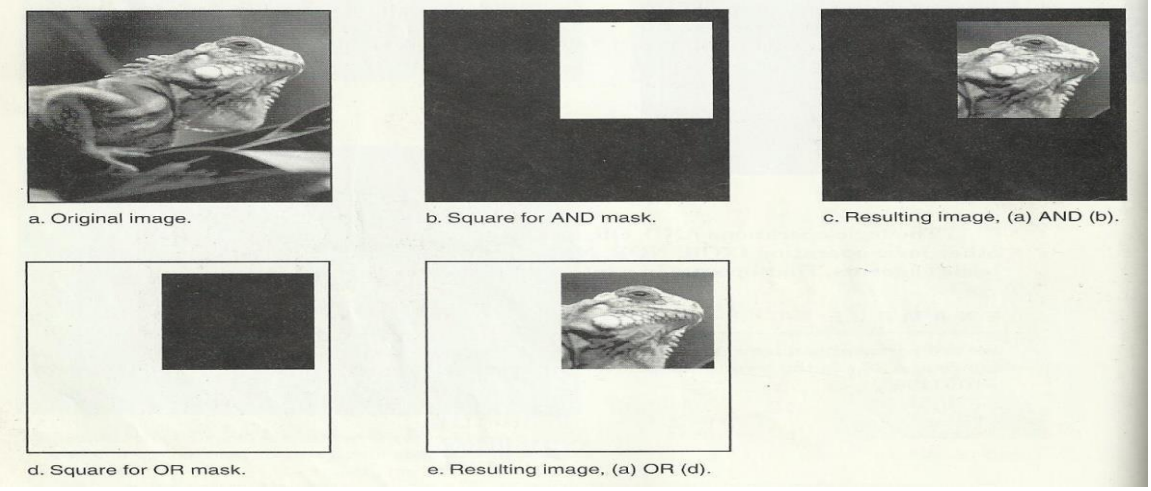


Figure (2.9): Image masking.

In addition to masking, logical operation can be used in feature detection. Logical operation can be used to compare between two images, as shown below:

- **AND**

This operation can be used to find the ***similarity*** white regions of two different images (it required two images).

$$g(x,y) = a(x,y) \wedge b(x,y)$$

- **Exclusive OR**

This operator can be used to find the ***differences*** between white regions of two different images(it requires two images).

$$g(x,y) = a(x,y) \otimes b(x,y)$$

- **NOT**

This operator can be performed on grey-level images, it's applied on only one image, and the result of this operation is the ***negative*** of the original image.

g(x,y) =255- f (x,y)

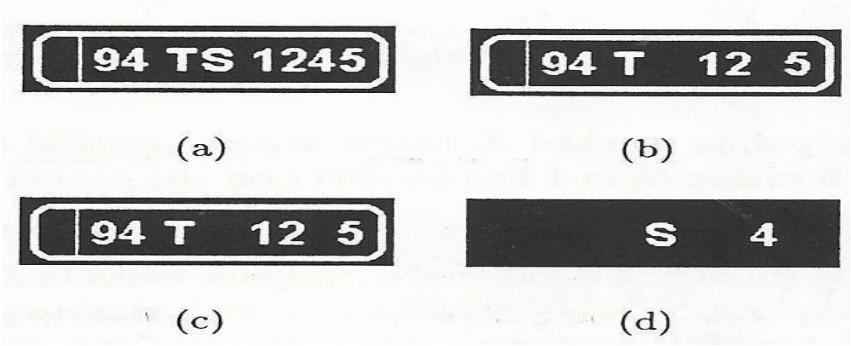
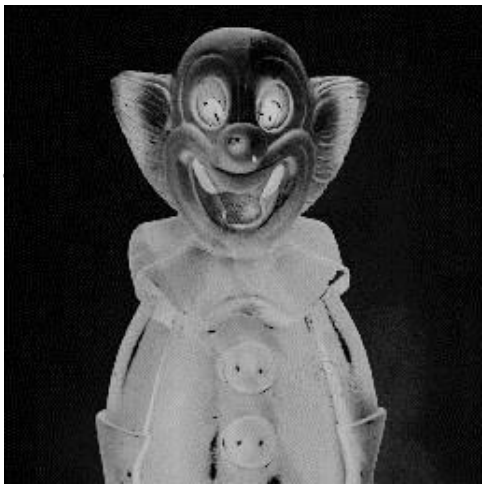


Figure 2.10:

(a) Input image a(x,y)
(b) Input image b(x,y) ,
(c) a(x,y)^ b(x,y),
(d) a(x,y)b(x,y) ⊗



a. Original image



b. Image after NOT operation.

Figure (2.11): Complement Image.

Example: A logic AND is performed on two images, suppose the two corresponding pixel values are $(111)_{10}$is one image and $(88)_{10}$ in the second image. The corresponding bit strings are:

$(111)_{10}$ ⟶ 01101111_2

AND

$(88)_{10}$ ⟶ 01011000_2

01001000

Example: Find the output image for the AND logical operation between the following sub images:

$$\boldsymbol{A} = \begin{bmatrix} 50_{10} & 11_{10} \\ 18_{10} & 22_{10} \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 10_{10} & 60_{10} \\ 33_{10} & 130_{10} \end{bmatrix}$$

Image Processing & Machine Vision

Chapter 1

Introduction to Computer Vision and Image Processing

1.1 Computer Imaging

Can be defined as acquisition and processing of visual information by computer. Computer representation of an image requires the equivalent of many thousands of words of data, so the massive amount of data required for image is a primary reason for the development of many sub areas with field of computer imaging, such as image compression and segmentation .Another important aspect of computer imaging involves the ultimate "receiver" of visual information in some case the human visual system and in some cases the human visual system and in others the computer itself.

Computer imagingcan be separate into two primary categories:

1. **Computer Vision.**
2. **Image Processing.**

(In computer vision application the processed images output for use by a computer, whereas in image processing applications the output images are for human consumption).

These two categories are not totally separate and distinct. The boundaries that separate the two are fuzzy, but this definition allows us to explore the differences between the two and to explore the difference between the two and to understand how they fit together (Figure 1.1).

Computer imaging can be separated into two different but overlapping areas.

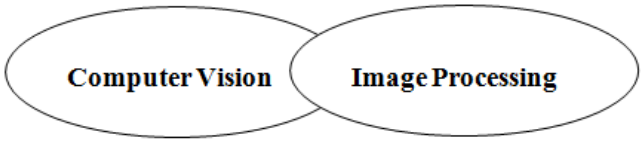


Figure (1.1): Computer Imaging [1].

Historically, the field of image processing grew from electrical engineering as an extension of the signal processing branch, whereas are the computer science discipline was largely responsible for developments in computer vision.

1.2 Computer Vision

Computer vision emulate human vision, that's mean: understanding the scene based on image data. One of the major topics within this field of computer vision is image analysis.

Image Analysis: involves the examination of the image data to facilitate solving vision problem.

The image analysis process involves two other topics:

- Feature Extraction: is the process of acquiring higher level image information, such as shape or color information.
- Pattern Classification: is the act of taking this higher –level information and identifying objects within the image.

Computer vision systems are used in many and various types of environments, such as:

1. Manufacturing Systems: computer vision is often used for quality control, where the computer vision system will scan manufactured items for defects, and provide control signals to a robotics manipulator to remove detective part automatically.

2. Medical Community: current example of medical systems to aid neurosurgeons during brain surgery, systems to diagnose skin tumors automatically.
3. The field of Law Enforcement and security in an active area for computer vision system development, with application ranging from automatic identification of fingerprints to DNA analysis.
4. Infrared Imaging
5. Satellites Orbiting

1.3 Image Processing

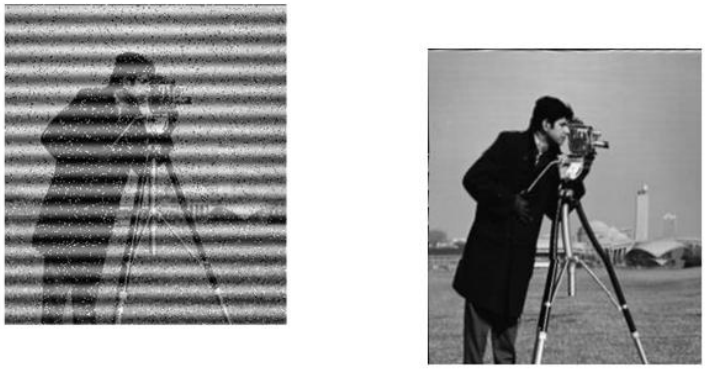
Image processing does some transformations on image. That means may be it does some smoothing,sharpening, contrasting,stretching on the image for making image more enhancive & readable that is input and output of a process are images.. In other words the image are to be examined and a acted upon by people.

The major topics within the field of image processing include:

1. Image restoration.
2. Image enhancement.
3. Image compression.

1.3.1 Image Restoration

Is the process of taking an image with some known, or estimated degradation, and restoring it to its original appearance. Image restoration is often used in the field of photography or publishing where an image was somehow degraded but needs to be improved before it can be printed(Figure 1.2).



a. Image with distortion

b. Restored image

Figure (1.2) Image Restoration

1.3.2 Image Enhancement

Involves taking an image and improving it visually, typically by taking advantages of human Visual Systems responses. One of the simplest enhancement techniques is to simply stretch the contrast of an image. Enhancement methods tend to be problem specific. For example, a method that is used to enhance satellite images may not suitable for enhancing medical images.

Although enhancement and restoration are similar in aim, to make an image look better.Restoration method attempt to model the distortion to the image and reverse the degradation, where enhancement methods use knowledge of the human visual systems responses to improve an image visually.



a. image with poor contrast



b. Image enhancement bycontrast

Figure (1.3) Image Enhancement

1.3.1 Image Compression

Involves reducing the typically massive amount of data needed to represent an image. This done by eliminating data that are visually unnecessary and by taking advantage of the redundancy that is inherent in most images. Image data can be reduced 10 to 50 times, and motion image data (video) can be reduced by factors of 100 or even 200.



Image before compression (92) KB



b. Image after compression - (6.59)KB

1.4 Computer Imaging Systems

Computer imaging systems are comprised of two primary components types, hardware and software. The hard ware components can be divided into image acquiring sub system (computer, scanner, and camera) and display devices (monitor, printer).The software allows us to manipulate the image and perform any desired processing on the image data.

1.5 Digitization

The process of transforming a standard video signal into digital image. This transformation is necessary because the standard video signal in analog (continuous) form and the computer requires a digitized or sampled version of that continuous signal. The analog video signal is turned into a digital image by sampling the continuous signal at affixed rate. In the figure below we see one line of a video signal being sampled (digitized) by instantaneously measuring the voltage of the signal (amplitude) at fixed intervals in time.

The value of the voltage at each instant is converted into a number that is stored, corresponding to the brightness of the image at that point. Note that **the image brightness of the image at that point** depends on both the intrinsic properties of the object and the lighting conditions in the scene.

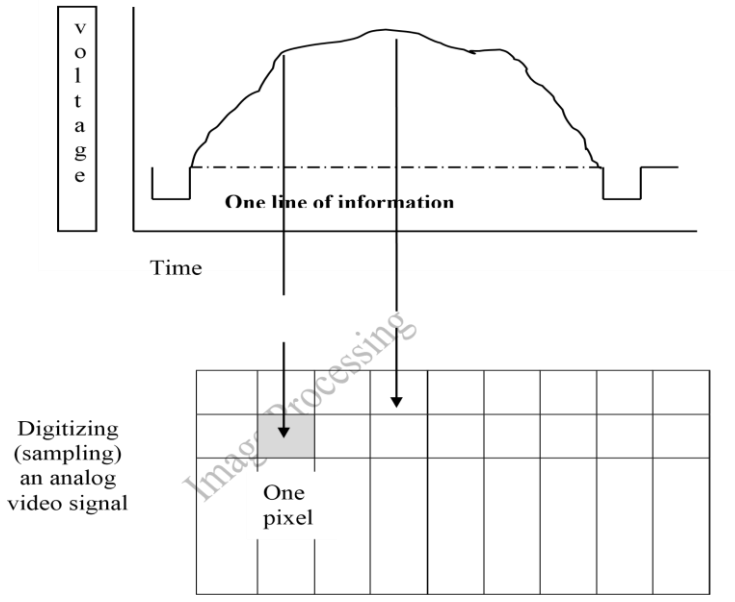


Figure (1.5) Digitizing (Sampling) an Analog Video Signal

- The image can now be accessed as a two-dimension array of data , where each data point is referred to a pixel (picture element).for digital images we will use the following notation :

- I(r,c) = The brightness of image at the point (r,c) Where r= row and c= column.
- “When we have the data in digital form, we can use the software to process the data”.
- The digital image is 2D- array as:

$$\begin{bmatrix} I(0,0) & I(0,1) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots I(0,N-1) \\ I(1,0) & I(1,1) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots I(1,N-1) \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ I(N-1,0) & I(N-1,1) \ldots\ldots\ldots\ldots\ldots\ldots I(N-1,N-1) \end{bmatrix}$$

In above image matrix, the image size is (N×N) [matrix dimension] then:

$$Ng = 2^{m}$$

Where Ng denotes the **number of gray** levels m, where m is the no. of bits contains in digital image matrix.

Example: If we have (6 bit) in 128 X 128 image .Find the no. of gray levels to represent it ,then find the no. of bit in this image?

Solution:

$N_g = 2^6 = 64$ Gray Level

$N_b = 128 * 128 * 6 = 9.8304 * 10^4$ bit

1.6 The Human Visual System

The Human Visual System (HVS) has two primary components:

- Eye.
- Brain.

* The structure that we know the most about is the image receiving sensors (the human eye).

* The brain can be thought as being an information processing unit analogous to the computer in our computer imaging system.

These two are connected by the optic nerve, which is really a bundle of nerves that contains the path ways for visual information to travel from the receiving sensor (the eye) to the processor (the brain).

1.7 Image Resolution

Pixels are the building blocks of every digital image. Clearly defined squares of light and color data are stacked up next to one another both horizontally and vertically. Each **picture element (pixel for short)** has a dark to light value from 0 (solid black) to 255 (pure white).

That is, there are 256 defined values. **A gradient** is the gradual transition from one value to another in sequence. At the heart of any convincing photographic rendering is a smooth, seamless and beautiful gradation.

In computers, resolution is the number of pixels (individual points of color) contained on a display monitor, expressed in terms of the number of pixels on the horizontal axis and the number on the vertical axis. The sharpness of the image on a display depends on the resolution and the size of the monitor. The same pixel resolution will be sharper on a smaller monitor and gradually lose sharpness on larger monitors because the same numbers of pixels are being spread out over a larger number of **inches**.

Display resolution is not measured in dots per inch as it usually is with printers *(We measure resolution in pixels per inch or more commonly, dots per inch (dpi)).*

- A display with 240 pixel columns and 320 pixel rows would generally be said to have a resolution of **240×320.**
  - Resolution can also be used to refer to the total number of pixels in a digital camera image. For example, a camera that can create images of 1600x1200 pixels will sometimes be referred to as a 2 megapixel resolution camera since 1600 x 1200 = 1,920,000 pixels, or roughly 2 million pixels. Where a megapixel (that is, a million pixels) is a unit of image sensing capacity in a digital camera. In general, the more megapixels in a camera, the better the resolution when printing an image in a given size.

- Below is an illustration of how the same image might appear at different pixel resolutions, if the pixels were poorly rendered as sharp squares (normally, a smooth image reconstruction from pixels would be preferred, but for illustration of pixels, the sharp squares make the point better).

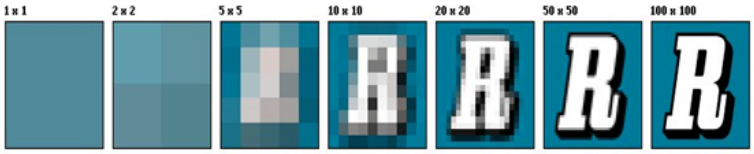


Figure (1.6) : Image Resolution.

- An image that is 2048 pixels in width and 1536 pixels in height has a total of 2048×1536 = 3,145,728 pixels or 3.1 megapixels. One could refer to it as 2048 by 1536 or a 3.1-megapixel image.

1.8 Image Brightness Adaption

- Brightness is intensity of light in simple words. Adaptation basically is "getting used to" and be comfortable with it. So conceptually brightness adaption is basically "getting used to changes in brightness/ changes intensity of light".
- A simple example is when you go out into light from darkness you take some time to "get used" to the brightness outside and feel comfortable. This is what exactly is brightness adaption.

- In image we observe many brightness levels and the vision system can adapt to a wide range.
- If the **mean value** of the pixels inside the image is around Zero gray level then the brightness is low and the images dark but for mean value near the 255 then the image is light.
- If fewer gray levels are used, we observe false contours lines resulting from gradually changing light intensity not being accurately represented.

- The human visual system HVS can perceive approximately 10^{10} different light intensity levels. However, at any one time we can only discriminate between a much smaller number – brightness adaptation.

1.9 Image Representation

- We have seen that the human visual system (HVS) receives an input image as a collection of spatially distributed light energy; this is form is called an optical image.
- Optical images are the type we deal with every day – cameras captures them, monitors display them, and we see them [we know that these optical images are represented as video information in the form of analog electrical signals and have seen how these are sampled to generate the digital image I(r , c).

- The digital image I (r, c) is represented as a two- dimensional array of data, where each pixel value corresponds to the brightness of the image at the point (r, c). in linear algebra terms , a two-dimensional array like our image model I(r, c) is referred to as a matrix , and one row (or column) is called a vector. The image types we will consider are:

1.Binary Image

- Binary images are the simplest type of images and can take on two values, typically black and white, or '0' and '1'. A binary image is referred to as a 1 bit/pixel image because it takes only 1 binary digit to represent each pixel.
- These types of images are most frequently in computer vision application where the only information required for the task is general shapes, or outlines information.
- For example, to position a robotics gripper to grasp an object or in optical character recognition (OCR).

- Binary images are often created from gray-scale images via a threshold value is, those values above it are turned white (‘1’), and those below it are turned black (‘0’).

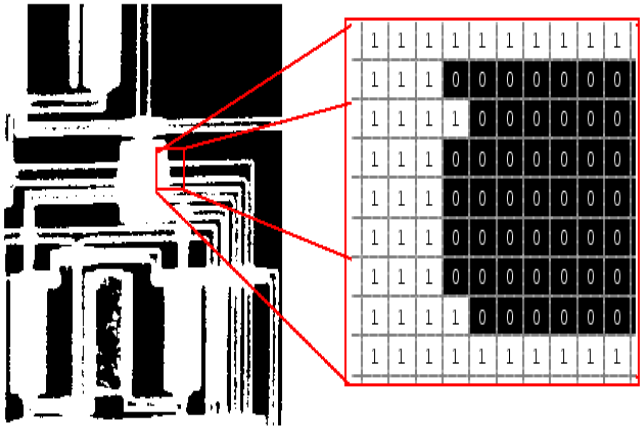


Figure (1.7): Binary Images.

2- Gray-scale images

- Gray-scale image is a range of monochromatic shades from black to white. Therefore, a grayscale image contains only shades of gray (brightness information only)and no color information.
- The number of different brightness level available. (0)value refers to black color, (255) value refers to white color, and all intermediate values are different shades of gray varying from black to white.

- The typical image contains 8 bit/ pixel (data, which allows us to have (0-255) different brightness (gray) levels.
- The 8 bit representation is typically due to the fact that the byte, which corresponds to 8-bit of data, is the standard small unit in the world of digital computer.



Figure (1.8) : Gray Scale Image

3- Color Images

- Color image can be modeled as three band monochrome image data, where each band of the data corresponds to a different color.

- The actual information stored in the digital image data is brightness information in each spectral band.

- When the image is displayed, the corresponding brightness information is displayed on the screen by picture elements that emit light energy corresponding to that particular color as shown in Figure (1.9).

- Typical color images are represented as red, green, and blue or RGB images .using the 8-bit monochrome standard as a model, the corresponding color image would have 24 bit/pixel – 8 bit for each color bands (red, green and blue).



Figure (1.9): Color Image

- The following figure illustrate that in addition to referring to a row or column as a vector, we can refer to a single pixel red ,green, and blue values as a color pixel vector –(R,G,B).

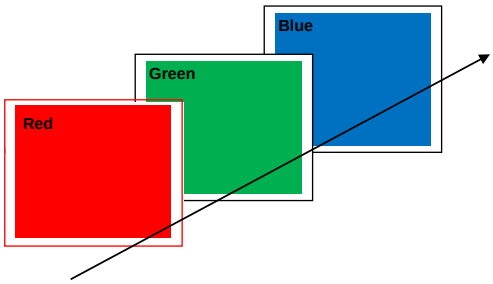


Figure (1.10) :A color pixel vector consists of the red, green and blue pixel values (R, G, B) at one given row/column pixel coordinate(r , c)

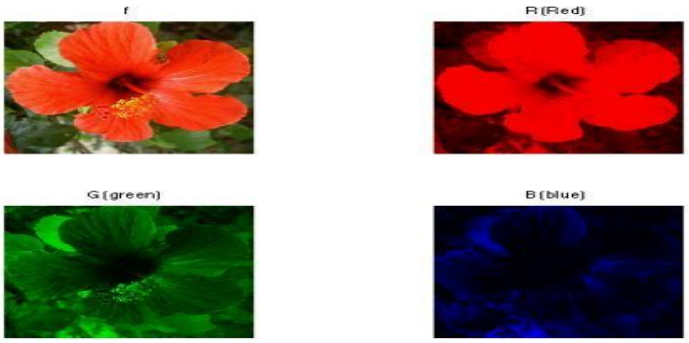


Figure (1. 11): A color pixel vector consists of the red, green and blue.

- For many applications, RGB color information is transformed into mathematical space that decouples the brightness information from the color information.
- The hue/saturation/lightness (HSL) color transform allows us to describe colors in terms that we can more readily understand.

- The lightness is the brightness of the color, and the hue is what we normally think of as “color” (ex: green, blue, red, and orange). The saturation is a measure of how much white is in the color (ex: Pink is red with more white, so it is less saturated than a pure red).

- **Example:**

  “a deep, bright orange” would have a large intensity (“bright”), a hue of “orange”, and a high value of saturation (“deep”).we can picture this color in our minds, but if we defined this color in terms of its RGB components, R=245, G=110 and B=20.
- Modeling the color information creates a more people oriented way of describing the colors.

4.Multispectral Images

A multispectral image is one that captures image data at specific frequencies across the electromagnetic spectrum. Multispectral images typically contain information outside the normal human perceptual range.

This may include infrared , ultraviolet , X-ray, acoustic or radar data. Source of these types of image include satellite systems, underwater sonar systems and medical diagnostics imaging systems.

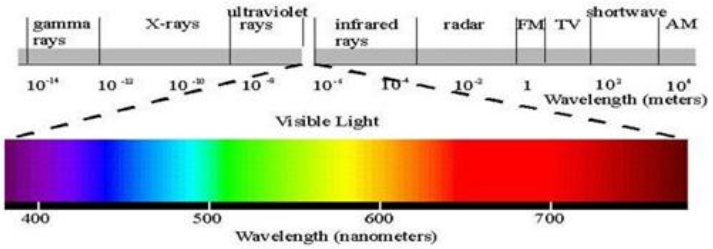


Figure (1.12): Electromagnetic spectrum.



Figure (1.13): Multispectral images

Computer Graphics

- Graphics are visual presentations on some surface, such as a wall, computer screen, paper, etc. Examples are photographs, drawings, graphs, diagrams, maps, engineering drawings, or other images
- **Computer graphics** are graphics created using computers and, more generally, the representation and manipulation of pictorial data by a computer.
- Developments in computer graphics have had a deep impact on many types of media and have revolutionized the animation and video game industry.

Computer Graphics

- The term Computer Graphics has several meanings:
  - the representation and manipulation of pictorial data by a computer
  - the various technologies used to create and manipulate such pictorial data.

What is an Image?

- Image is a source of information according to information theory
- An image may be defined as a two dimensional function f(x,y) where x and y are spatial coordinates and amplitude of f at any pair of coordinates (x,y) is called the intensity or Gray level of the image at that point.

Digital Image

- When x,y and the amplitude values of f are all finite, discrete quantities, we call the image a Digital Image.
- A digital Image is composed of a finite number of elements each of which has a particular location and value
- These elements are referred to as Picture Elements, Image Elements, or Pixels.

Pixel

- In digital imaging, a pixel is the smallest piece of information in an image.
- Pixels are normally arranged in a regular 2-dimensional grid, and are often represented using dots or squares
- The intensity of each pixel is variable; in color systems, each pixel has typically three or four components such as red, green, and blue, or cyan, magenta, yellow, and black

Why do we Process Images?

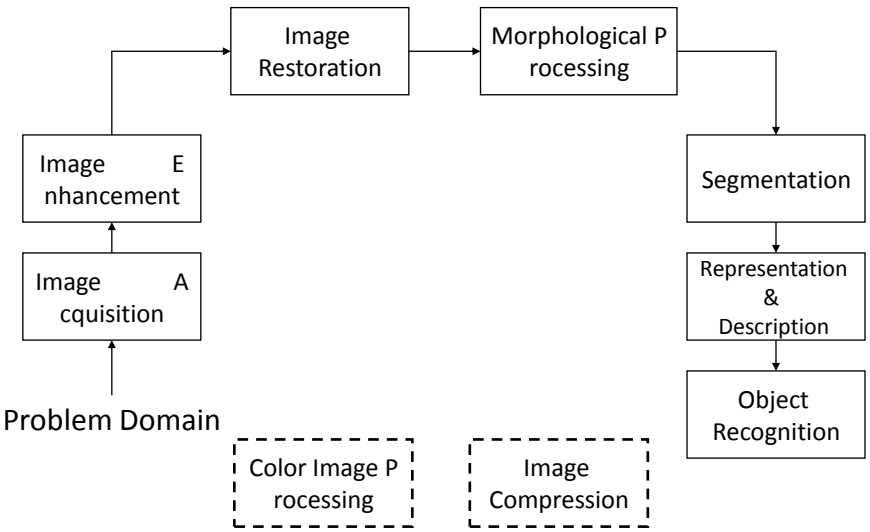
Facilitate picture storage and transmission

Enhance and restore images

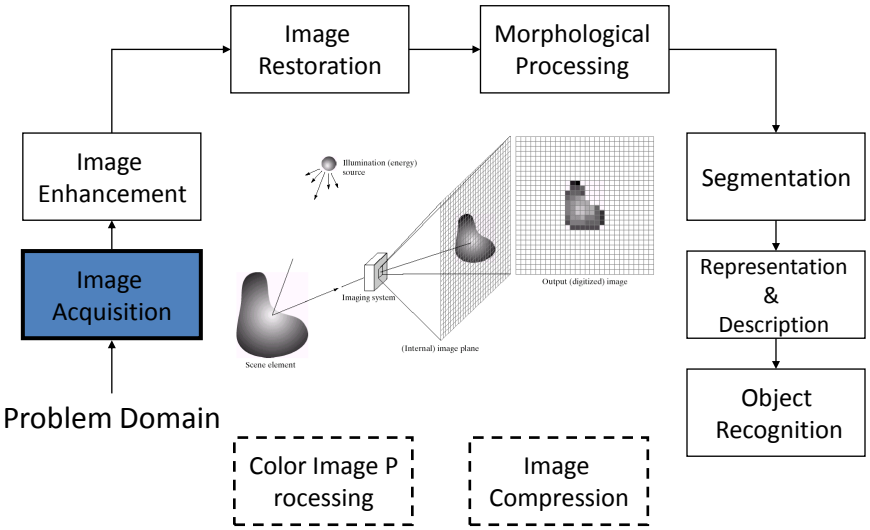
Extract information from images

Prepare for display or Printing

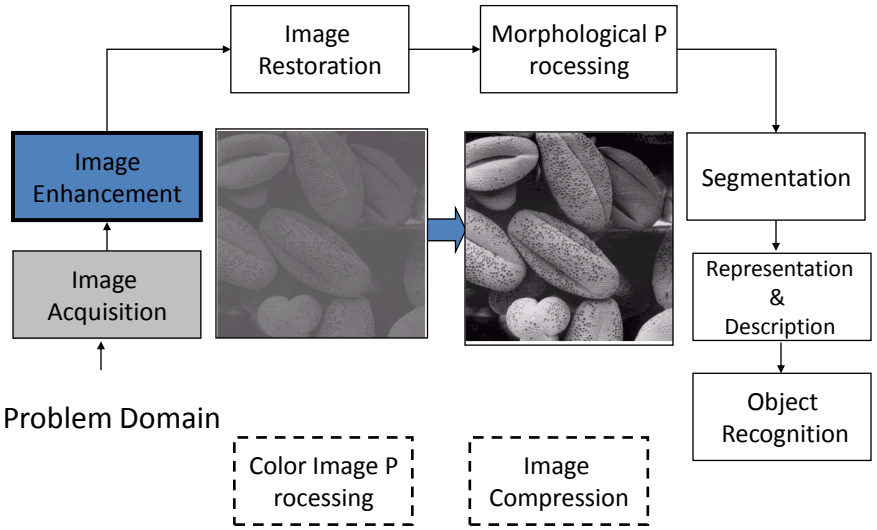
Key Stages in Digital Image Processing



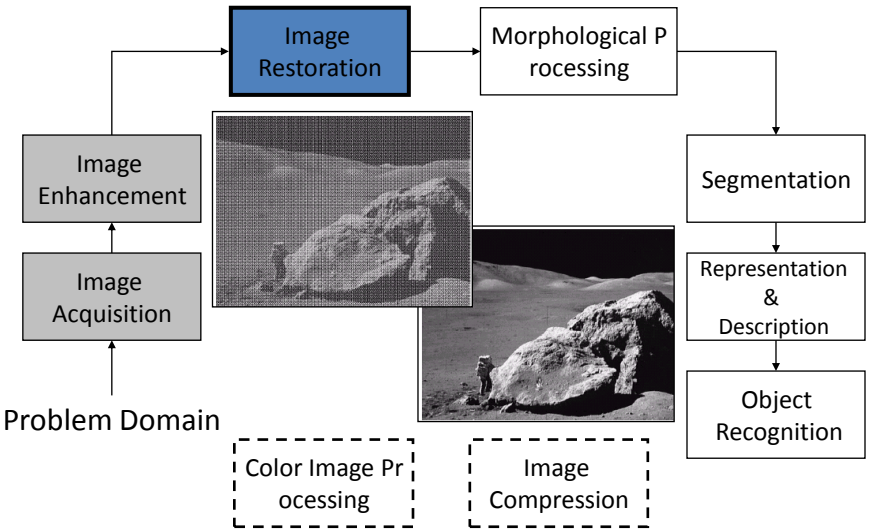
Key Stages in Digital Image Processing: Image Acquisition



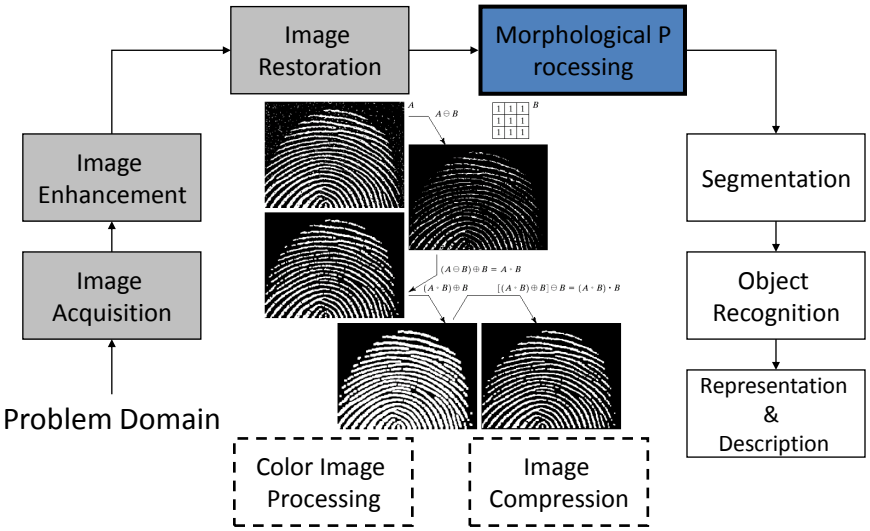
Key Stages in Digital Image Processing: Image Enhancement



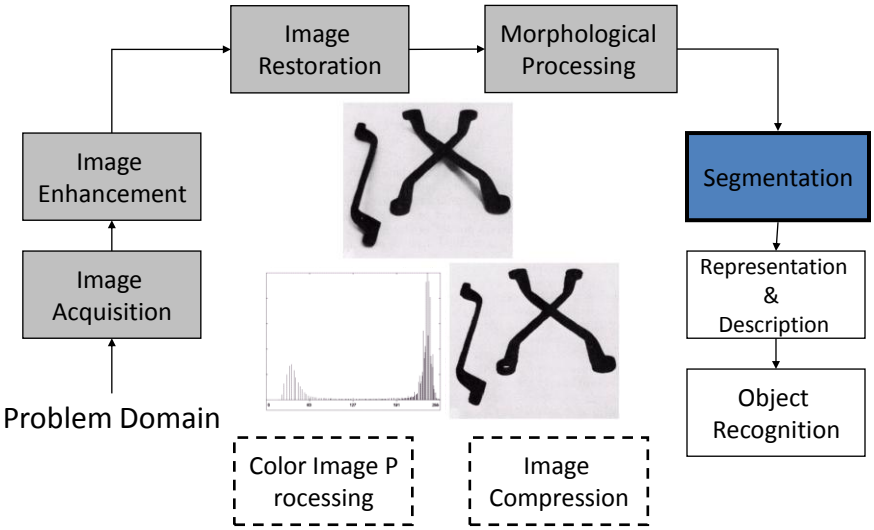
Key Stages in Digital Image Processing: Image Restoration



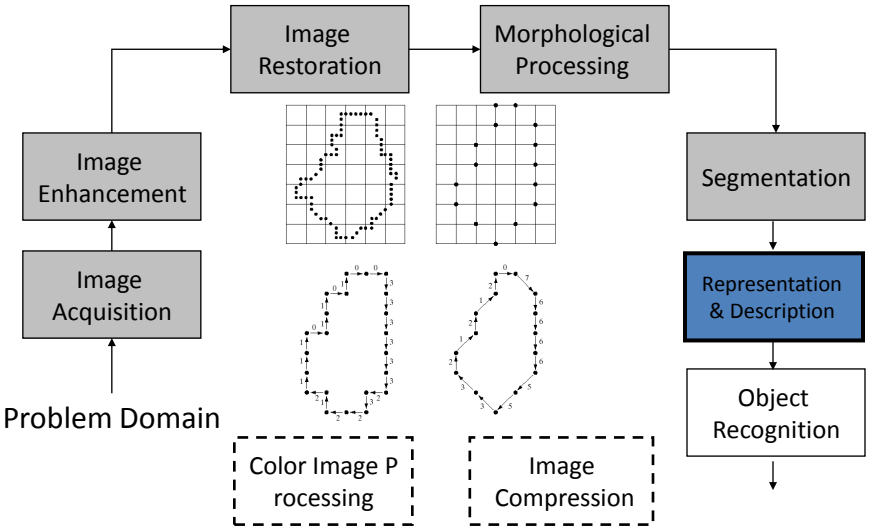
Key Stages in Digital Image Processing: Morphological Processing



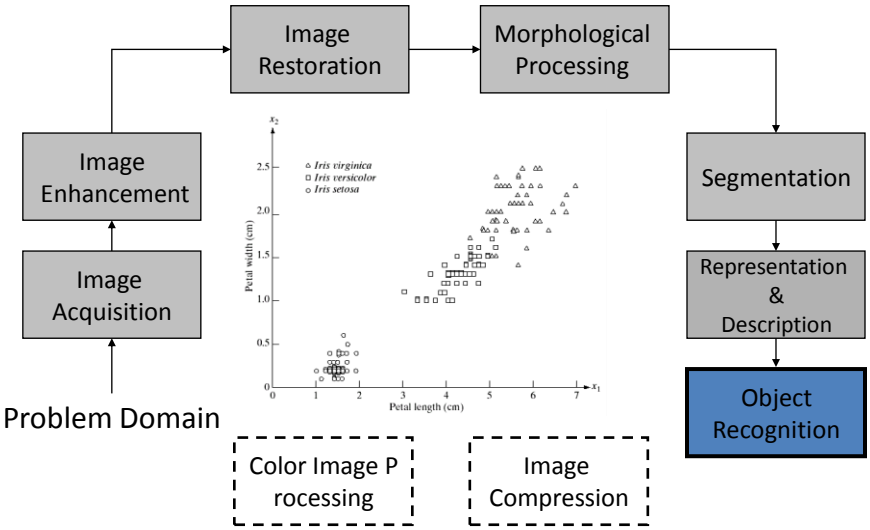
Key Stages in Digital Image Processing: Segmentation



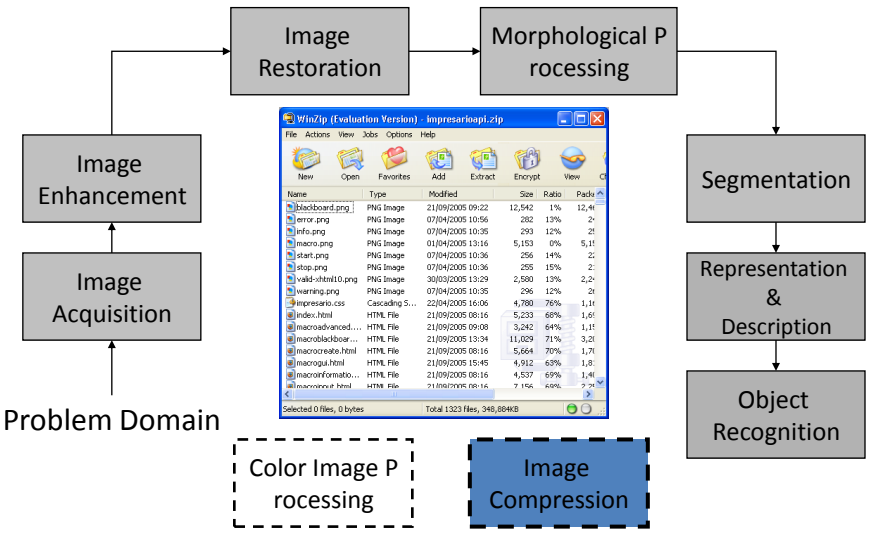
Key Stages in Digital Image Processing: Representation & Description



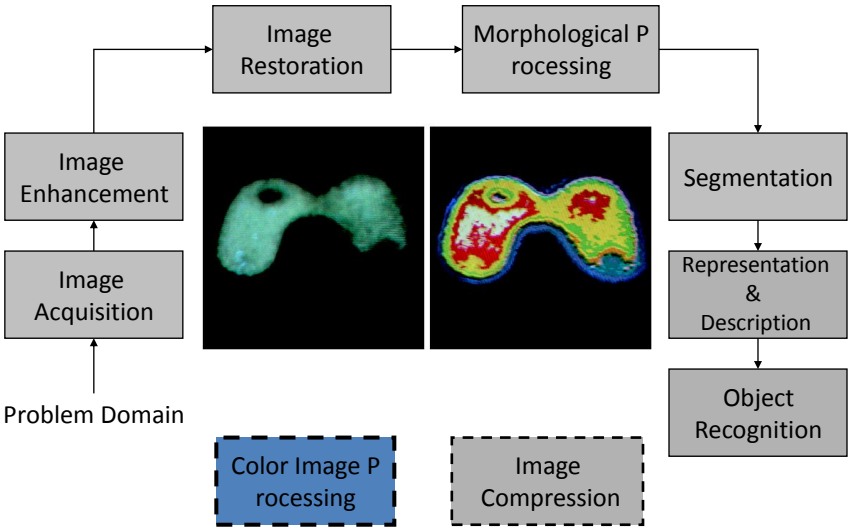
Key Stages in Digital Image Processing: Object Recognition



Key Stages in Digital Image Processing: Image Compression



Key Stages in Digital Image Processing: Colour Image Processing



Applications of Image Processing

1. Image Restoration



Damaged Image



Restored Image

Image Colorization



Original B/W Image



Colorized Image



Original Image



Colorized Image

Image Enhancement



Original Images Enhanced Images

Face Detection



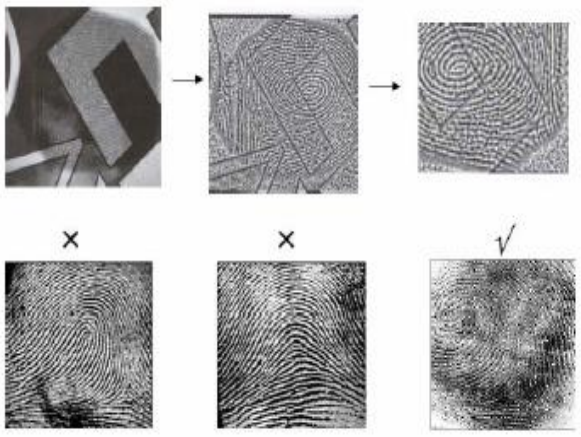
Face Tracking



Face Morphing



Finger Print Recognition



Personal Identification Using Iris Recognition



Image Processing Components

Image Sensing device

Storage Media

Processing Systems

Displays

Communication Media

Hardcopy devices (e.g Printer)